Name: Date: Period:

# Problem Solving Method Practice

## Equations

$$\bar{s} = \frac{d}{\Delta t} \qquad \bar{v} = \frac{\Delta x}{\Delta t} \qquad \Delta x = x_f - x_0$$

$$1\,\mathrm{m/s} = 3.6\,\mathrm{km/h}$$

## Problem Solving Method

1. Picture
2. Knowns/Unknowns
3. Check units
4. Pick an equation (solve algebraically)
5. Plug & Chug
6. Answer with units

## Problems

1. A certain object has a velocity of 25 km/h.
   (a) How much time will it take to travel 150 m?
   (b) How about 360 m?

2. An airplane travels 2100 km at a velocity of 720 km/h and then encounters a tailwind that boosts its speed to 990 km/h for the next 2800 km.
   (a) What was the total time of the trip?
   (b) what was the airplane's average velocity?

3. Two bicycles start from the same point, but one is ahead of the other. The front bicycle travels at a constant speed of 12 m/s, while the back bicycle travels at a constant speed of 18 m/s. The initial distance between the two bicycles is 150 meters. Calculate where the back bicycle overtakes the front bicycle.